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abstract

I. INTRODUCTION

Gravitational waves provide a unique observational window into astrophysical and cosmological phenomena. The first direct detection in 2015 established gravitational-wave observations of compact binary mergers with ground-based interferometers [1]. Accessing gravitational waves at millihertz frequencies requires a space-based detector to overcome terrestrial seismic and Newtonian noise and to provide sufficiently long interferometric baselines. The Laser Interferometer Space Antenna (LISA), a space-based observatory led by ESA with NASA participation, is designed to operate in the 0.1 mHz to 1 Hz band [2].

The LISA constellation consists of three spacecraft arranged in an approximately equilateral triangle with arm lengths of about 2.5 million km. Laser beams exchanged between the spacecraft form a constellation-scale interferometer. Unlike ground-based detectors with effectively equal arms, the LISA arm lengths vary by up to about 1.2% due to orbital motion, preventing direct cancellation of laser frequency noise. Although cavity pre-stabilisation reduces laser frequency noise substantially, it remains several orders of magnitude above the displacement noise budget required for gravitational-wave detection [3]. To suppress this noise, time delay interferometry (TDI) has been developed as a post-processing technique that synthesizes virtual equal-arm interferometers by appropriately time-shifting and combining phasemeter measurements.

In addition to laser frequency noise, LISA measurements are affected by clock noise originating from the independent ultra-stable oscillators (USOs) carried by each spacecraft. Since all phase measurements are referenced to these clocks, timing jitter couples directly into the measured heterodyne beatnotes. Space-qualified USOs achieve Allan deviations of order 10^{-13} at one-second averaging times [4], which is insufficient for the picometer-level test-mass position sensitivity required by LISA. To mitigate this effect, LISA employs additional clock-noise transfer measurements—implemented via modulation sidebands exchanged between spacecraft.

TDI has been developed analytically and tested through numerical simulations [3, 5]. Simulations predict the response of a specified measurement model and test its numerical implementation. Physical experiments provide a complementary test of whether the hardware follows that model, including couplings and processing effects that an idealized simulation may omit.

The University of Florida Laser Interferometry Simu-

lator (UFLIS) demonstrated TDI and TDI-ranging with time-varying electronic propagation delays and optical laser signals [6]. miniLISA continues this approach with FPGA delay lines and independent spacecraft clock references. The experiment of de Vine et al. demonstrated laser-noise suppression of approximately 10^9 and clock-noise suppression of 6×10^4 , using short optical paths and a negligible-arm-delay approximation [7]. LISA On Table demonstrated first-generation TDI with simulated noise and static unequal arms [8].

Here, we present an overview of an ongoing effort to develop a TDI testbed that combines realistic laser frequency noise, programmable delay lines, and independent clock sources within a single experimental framework. The testbed focuses on reproducing the dominant noise couplings of the LISA metrology system—specifically, laser frequency noise and clock noise—while preserving the structure imposed by unequal and time-dependent delays. Crucially, the same delay lines that impose the inter-spacecraft light-travel times also allow physically consistent gravitational-wave strain signals to be injected directly into the measurement chain. The testbed can therefore be used not only to demonstrate noise suppression, but to verify that a known gravitational-wave signal survives TDI processing of hardware-generated data with the expected amplitude and phase.

An initial purely electrical implementation centered around field programmable gate array (FPGA)-based digital delay lines has already been demonstrated in Ferguson et al. [9]. The developments discussed here extend that architecture by incorporating an optical front-end with physically representative laser noise and independently running spacecraft clocks. This reproduces the laser-noise and propagation structure of the one-way measurements together with a reduced clock-transfer model, allowing TDI combinations and clock-noise calibration to be formed entirely from hardware-generated data. The goal of this work is to realize a laboratory-scale analog of the LISA inter-satellite metrology system—a “miniLISA”—that bridges the gap between idealized software simulations and the flight instrument, and provides a controlled environment for testing the robustness of TDI to instrumental non-idealities and realistic gravitational-wave signal injection.

More concretely, we look for residual laser and clock noise below the secondary-noise floor after processing, together with recovery of an injected GW signal at the expected amplitude and phase.

Section II introduces the LISA measurements and

noise-suppression combinations. Section III describes their laboratory implementation, and Section IV estimates its performance from measured and calculated component noises. We then discuss possible extensions and remaining measurements.

II. LISA INTERFEROMETRIC DETECTION SYSTEM

We now describe the LISA interferometric detection system (IDS) and present a mathematical model of the noises. We specifically focus on relevant elements for laser frequency noise and USO jitter propagation and coupling.

A. Interferometric Phase Measurements

Each of the three identical LISA satellites carries two lasers on board. We refer to one as the local laser and the other as the adjacent laser. These two laser fields interfere on the test-mass and reference interferometers on the two optical benches of the spacecraft. In addition, the local laser interferes with the laser received from a distant spacecraft on the science interferometer. It is this science interferometer that encodes the gravitational-wave signal by measuring fluctuations in the spacecraft-to-spacecraft distance, while the test mass interferometer will measure the spacecraft-to-test mass displacement. In Figure 1 (a), a simplified LISA two-way armlink involving only the science interferometer is shown.

We begin by writing the optical phase of the laser field originating from movable optical system assembly (MOSA) ij on spacecraft i facing spacecraft j as a function of time t :

$$\Phi_{ij}(t) = \nu_{ij}t + \delta\phi_{ij}(t), \quad (1)$$

where ν_{ij} denotes the nominal carrier frequency and $\delta\phi_{ij}(t)$ represents the intrinsic laser phase noise. Throughout this section, phases are expressed in cycles, frequencies in Hz, and clock errors in seconds. Capital Φ denotes total phase, while $\delta\phi$ denotes phase fluctuations about the nominal linear evolution. We use a single time coordinate, neglect relativistic time transformations, and retain constant nominal frequencies in the noise-coupling coefficients. The clock on spacecraft i reads $t_i = t + q_i(t)$, where q_i is its timing error.

Each of the laser fields is phase-modulated using an electro-optic modulator (EOM) driven by the local USO. This is used to imprint sidebands on the laser field that carry the USO timing noise $q_i(t)$, creating a phase noise term $\nu_{ij}^m q_i(t)$. Since there is only one USO per spacecraft, clock noise carries a single spacecraft index.

The modulation frequency ν_{ij}^m will be about 2.4 GHz, which is upconverted from the USO via a phase locked loop (PLL) synthesizer, preserving its noise with high fidelity. For the adjacent and opposing MOSAs ik and

ji respectively, the corresponding modulation frequency differs by 1 MHz, so 2.401 GHz, such that the sideband-sideband beat notes are offset by 1 MHz from the carrier-carrier beat note. The modulation depth $\mu_{\text{EOM}} = 0.53 \text{ rad}$ places about 87% of the optical power in the carrier and 6.5% in each first-order sideband. Figure 2 shows how the USO signal is distributed to the modulation, sampling-clock, and pilot-tone synthesis chains. Their role in clock-noise suppression is discussed in subsection II C. Additional phase modulation with pseudo-random codes supports ranging and data transfer; it is distinct from the sinusoidal clock-transfer modulation considered here [10].

Propagation from MOSA ij to ji delays the emitted phase and adds the GW response along the arm. Following the separation of background propagation and GW perturbations in [3], we write

$$d_{ji}^{\text{tot}}(t) = d_{ji}(t) + H_{ji}(t). \quad (2)$$

Here $d_{ji}(t)$ is the background propagation time, which can vary slowly with the constellation breathing, and $H_{ji}(t)$ is its GW-induced perturbation. The first index denotes the receiving spacecraft and the second the emitting spacecraft. We define the delay operator using only the background propagation time,

$$\mathbf{D}_{ji}x(t) = x(t - d_{ji}(t)). \quad (3)$$

Applying the total propagation time to Eq. (1) and expanding to first order gives

$$\begin{aligned} \Phi_{ij}(t - d_{ji}(t) - H_{ji}(t)) &\simeq \nu_{ij}[t - d_{ji}(t)] \\ &\quad + \mathbf{D}_{ji}\delta\phi_{ij}(t) - \nu_{ij}H_{ji}(t), \end{aligned}$$

where the product $H_{ji}\mathbf{D}_{ji}\delta\phi_{ij}$ of the GW perturbation and laser-frequency noise has been neglected. We therefore introduce the GW phase contribution

$$\delta\phi_{ji}^{\text{GW}}(t) = -\nu_{ij}H_{ji}(t). \quad (4)$$

To this order, the received total phase is $\mathbf{D}_{ji}\Phi_{ij}(t) + \delta\phi_{ji}^{\text{GW}}(t)$: the GW signal is an additive phase contribution, separate from the background delay operator. For constant ν_{ij} , the corresponding frequency fluctuation is $\dot{\delta\phi}_{ji}^{\text{GW}} = -\nu_{ij}\dot{H}_{ji}$.

At spacecraft j , the received field i interferes with the local laser field of phase $\Phi_{ji}(t)$ on a photodetector. The resulting heterodyne signal contains both the carrier-carrier beat note and the sideband-sideband beat notes. Additional cross terms lie outside the heterodyne range of the phasemeter; a representative LISA frequency plan uses absolute beat frequencies from 5 to 25 MHz [11].

The electrical signal from the photodetector is then digitized by the phase measurement system (PMS) front-end electronics. The sampling process introduces a coupling of the local USO timing error $q_j(t)$ into the measured phase. For all measurement readouts, we adopt

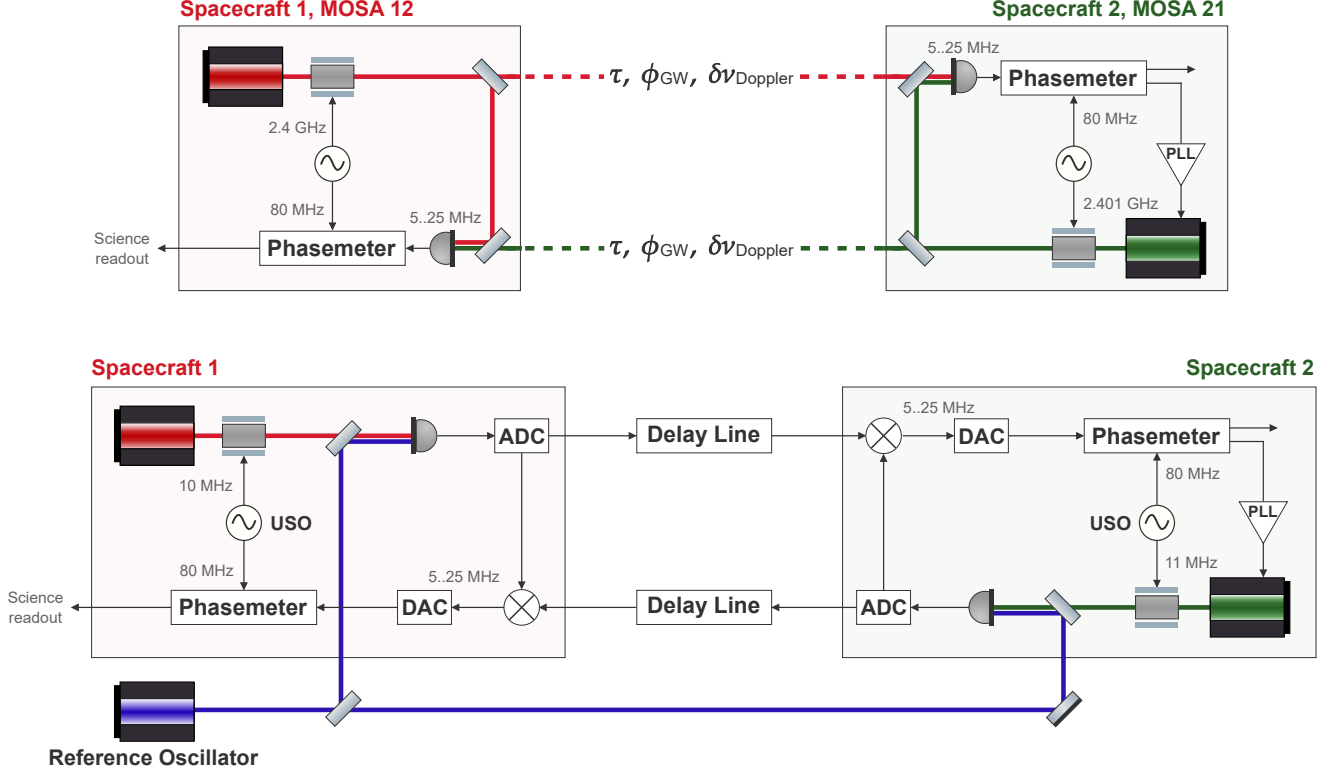


FIG. 1. A comparison of the two-way links as implemented in LISA (upper) and the testbed (lower). The upper panel shows the conceptual LISA configuration: the laser on Spacecraft 1 (MOSA 12) is phase-modulated at approximately 2.4 GHz using the local USO and transmitted to Spacecraft 2 (MOSA 21). The received beam interferes with the local laser, producing heterodyne beat notes in the MHz range that contain gravitational-wave phase fluctuations, Doppler shifts due to orbital motion, and laser frequency noise. These beat notes are tracked by the phasemeter, which is clocked by the onboard 80 MHz sampling reference derived from the USO. The lower panel depicts the laboratory implementation of the same signal topology. Instead of physical light propagation over millions of kilometres, the inter-spacecraft delay is emulated electronically using digital delay lines between ADC and DAC stages. A common reference oscillator is used to downconvert the laser fields from hundreds of terahertz to tens of megahertz, which can then be sampled by the delay line ADCs. The actual LISA-like heterodyne beat notes are created in the mixers.

the LISA convention that phases are written as “distant minus local.”¹ We define the nominal carrier coefficient

$$\alpha_{ji} = \Delta\nu_{ji} = \nu_{ij} - \nu_{ji}. \quad (5)$$

After removing the nominal beat evolution, the sampled carrier phase fluctuations are

$$\begin{aligned} \delta\phi_{ji}^c(t) = & \mathbf{D}_{ji}\delta\phi_{ij}(t) - \delta\phi_{ji}(t) \\ & + \delta\phi_{ji}^{\text{GW}}(t) - \alpha_{ji}q_j(t). \end{aligned} \quad (6)$$

Here α_{ji} is the beat frequency in the fixed-frequency approximation used for the noise model. For an orbit-dependent frequency plan, it would be replaced by the actual signed beat frequency, including Doppler shifts.

The signal is then passed to digital phase lock loops (DPLLs), which track the carrier and both sideband beat notes. In the following readouts, the nominal phase evolution is largely removed from the tracked phase, leaving the fluctuations used for GW detection and clock-noise suppression. To connect this noise model to the measurements used for TDI, we also include the secondary disturbances in the science readout:

$$\begin{aligned} \text{sci}_{ji}^c(t) = & \mathbf{D}_{ji}\delta\phi_{ij}(t) - \delta\phi_{ji}(t) - \alpha_{ji}q_j(t) \\ & + \delta\phi_{ji}^{\text{GW}}(t) + N_{ji}^c(t) \\ & - \frac{\nu_0}{c} (\mathbf{n}_{ij} \cdot \mathbf{D}_{ji}\Delta_{ij}(t) + \mathbf{n}_{ji} \cdot \Delta_{ji}(t)), \end{aligned} \quad (7)$$

The term $N_{ji}^c(t)$ represents combined readout noise from optical path-length fluctuations and electronics. The vectors $\Delta_{ij}(t)$ are optical-bench displacements in metres. The unit vector $\mathbf{n}_{ji}$ points along the incoming beam from spacecraft i to spacecraft j , and $\mathbf{n}_{ij} = -\mathbf{n}_{ji}$ in the ge-

¹ The phasemeter measures the absolute beat frequency; its sign is assigned using the frequency plan.

ometry used for these displacement terms. The factor $\nu_0/c = 1/\lambda_0$ converts path-length fluctuations to phase in cycles, with $\nu_0 \simeq 281$ THz for the nominal wavelength $\lambda_0 = 1064$ nm. MHz-scale offsets between lasers are small compared with this optical frequency, so a common ν_0 suffices for the displacement terms. Those offsets remain essential in the heterodyne clock coefficients, which depend on frequency differences.

The corresponding upper sideband-sideband beat note readout can be written as

$$\begin{aligned} \text{sci}_{ji}^{\text{sb},+}(t) = & \mathbf{D}_{ji} \delta \phi_{ij}(t) - \delta \phi_{ji}(t) - \gamma_{ji} q_j(t) \\ & + \delta \phi_{ji}^{\text{GW}}(t) + N_{ji}^{\text{sb},+}(t) \\ & - \frac{\nu_0}{c} (\mathbf{n}_{ij} \cdot \mathbf{D}_{ji} \Delta_{ij}(t) + \mathbf{n}_{ji} \cdot \Delta_{ji}(t)) \\ & + \nu_{ij}^m \mathbf{D}_{ji} q_i(t) - \nu_{ji}^m q_j(t). \end{aligned} \quad (8)$$

The sideband clock coupling coefficient is

$$\gamma_{ji} = \Delta \nu_{ji} + \nu_{ij}^m - \nu_{ji}^m = \Delta \nu_{ji} + \Delta \nu_{ji}^m = \Delta \nu_{ji}^{\text{sb},+}. \quad (9)$$

Again, the term $N_{ji}^{\text{sb},+}(t)$ represents stochastic readout noise specific to the upper sideband measurement. The lower sideband has the same laser phase noise but the opposite sign for each modulation contribution. The sidebands have optical frequencies $\nu_{ij} \pm \nu_{ij}^m$, so the same path-length change produces a slightly different phase shift in each sideband. Using ν_0/c for their displacement couplings neglects a fractional correction of order $\nu^m/\nu_0 \simeq 9 \times 10^{-6}$. Similarly, the upper-sideband GW phase is $-(\nu_{ij} + \nu_{ij}^m)H_{ji}$, differing from the carrier by $-\nu_{ij}^m H_{ji}$. We neglect this small difference in the readout equations above and use the same $\delta \phi_{ji}^{\text{GW}}$ for the carrier and both sidebands. The δ denotes a perturbation to the nominal optical phase, rather than stochastic noise.

For completeness, we also write out the reference and test mass interferometer readings on MOSA ji

$$\begin{aligned} \text{tm}_{ji}(t) = & \delta \phi_{jk}(t) - \delta \phi_{ji}(t) + \beta_{ji} q_j(t) + \mu_j(t) + N_{ji}^{\text{tm}}(t) \\ & + \frac{2\nu_0}{c} \mathbf{n}_{ji} \cdot (\delta_{ji}(t) - \Delta_{ji}(t)), \end{aligned} \quad (10)$$

$$\begin{aligned} \text{ref}_{ji}(t) = & \delta \phi_{jk}(t) - \delta \phi_{ji}(t) + \beta_{ji} q_j(t) \\ & + \mu_j(t) + N_{ji}^{\text{ref}}(t). \end{aligned} \quad (11)$$

Here we have introduced the fiber noise μ_j coming from the optical fiber that exchanges the local and adjacent laser fields between MOSA ji and MOSA jk . This term is typically referred to as the backlink noise. We have assumed it to be independent of the direction of propagation of the optical beams within. The vector δ_{ji} is the test-mass displacement in metres, including displacement produced by acceleration noise; it is not itself an acceleration. With the same sampling convention, the reference and test-mass clock coefficient is

$$\beta_{ji} = \nu_{ji} - \nu_{jk}. \quad (12)$$

It has the opposite sign to the adjacent-minus-local beat frequency, which explains the plus sign multiplying $\beta_{ji} q_j$ in the readouts.

These equations retain the terms needed to explain laser- and clock-noise suppression. A full instrument model includes further effects, such as tilt-to-length coupling and additional stray-light and optical-path disturbances. The miniLISA model omits the test-mass and reference interferometers and their associated bench-motion, backlink, and test-mass noise terms.

B. Laser Noise and TDI

In the LISA measurement system, laser frequency noise is first reduced onboard by locking one laser to an optical resonator and phase-locking the remaining lasers to this pre-stabilized reference with fixed frequency offsets. The last step ensures that all heterodyne beat notes remain within the phasemeter bandwidth while maintaining coherent phase relationships between spacecraft. However, residual laser frequency noise remains many orders of magnitude above the displacement sensitivity requirement and must be suppressed in post-processing. TDI achieves this by forming linear combinations of one-way phase measurements and appropriately delayed copies thereof, effectively synthesizing an equal-arm interferometer in post-processing.

For scale, consider a representative pre-stabilized frequency-noise amplitude of $30 \text{ Hz}/\sqrt{\text{Hz}}$ [3]. At Fourier frequency $f = 10 \text{ mHz}$, the corresponding single-laser phase-noise amplitude is $30/(2\pi f) \simeq 480 \text{ cycles}/\sqrt{\text{Hz}}$, equivalent to about $0.5 \text{ mm}/\sqrt{\text{Hz}}$ displacement noise at 1064 nm . This illustrative conversion is many orders of magnitude above picometre-scale displacement noise required for GW detection.

The construction proceeds in two steps: local measurements first remove optical-bench motion and reduce the six laser noises to three; delayed combinations of the resulting links then suppress the remaining laser noise [12, 13]. Using Eqs. (7), (10), and (11), we form

$$\begin{aligned} T_{ji} &= \text{tm}_{ji} - \text{ref}_{ji}, \\ \xi_{ji} &= \text{sci}_{ji}^c - \frac{1}{2} (T_{ji} + \mathbf{D}_{ji} T_{ij}). \end{aligned}$$

Within the common-optical-frequency approximation above, the bench terms cancel, leaving the test-mass contribution

$$M_{ji} = -\frac{\nu_0}{c} [\mathbf{n}_{ji} \cdot \delta_{ji}(t) + \mathbf{D}_{ji} (\mathbf{n}_{ij} \cdot \delta_{ij}(t))].$$

The following equations omit readout noises. Reciprocal backlink noise cancels in the local reference difference

$$z_{ji} = \frac{1}{2} (\text{ref}_{ji} - \text{ref}_{jk}) = \delta \phi_{jk} - \delta \phi_{ji} + \beta_{ji} q_j.$$

For cyclic triples $(i, j, k) = (1, 2, 3), (2, 3, 1), (3, 1, 2)$, the

intermediate variables can then be defined as

$$\begin{aligned}
\eta_{ji} &= \xi_{ji} - z_{ji} \\
&= \mathbf{D}_{ji}\delta\phi_{ij}(t) - \delta\phi_{jk}(t) - (\alpha_{ji} + \beta_{ji})q_j(t) \\
&\quad + \delta\phi_{ji}^{\text{GW}}(t) + M_{ji}(t), \\
\eta_{ij} &= \xi_{ij} + \mathbf{D}_{ij}z_{ji} \\
&= \mathbf{D}_{ij}\delta\phi_{jk}(t) - \delta\phi_{ij}(t) - \alpha_{ij}q_i(t) \\
&\quad + \mathbf{D}_{ij}[\beta_{ji}q_j(t)] \\
&\quad + \delta\phi_{ij}^{\text{GW}}(t) + M_{ij}(t).
\end{aligned}$$

Here M_{ij} is obtained by exchanging i and j in M_{ji} . These combinations reduce the number of independent laser phase noises from six to three, retaining $\delta\phi_{12}$, $\delta\phi_{23}$ and $\delta\phi_{31}$, while removing the optical-bench displacement terms. The construction introduces an asymmetry between the two link orientations.

For the two-arm geometry used below, we use the Michelson observable X_i . Define the synthesized round-trip measurements

$$R_{ij} = \eta_{ij} + \mathcal{D}_{ij}\eta_{ji}, \quad R_{ik} = \eta_{ik} + \mathcal{D}_{ik}\eta_{ki}.$$

The first-generation combination is

$$X_i = (1 - \mathcal{D}_{ij}\mathcal{D}_{ji})R_{ik} - (1 - \mathcal{D}_{ik}\mathcal{D}_{ki})R_{ij}. \quad (13)$$

Here $\mathbf{D}$ describes propagation in the measurement chain, while $\mathcal{D}$ denotes a delay applied in post-processing. Cancellation requires the processing delays to match the physical delays. A mismatch δd leaves a phase residual proportional to $\delta d \delta\phi$, so either an inaccurate delay estimate or an untracked change in the physical delay can limit laser-noise suppression.

First-generation TDI combinations cancel laser noise for constant light-travel times. Orbital motion makes these delays time-dependent, so their order of application matters and the cancellation is no longer exact. Second-generation combinations use longer sequences of delays to cancel the leading residual from arm-length changes, leaving terms involving higher powers of the arm velocities and arm accelerations [13].

C. Clock-noise Coupling and Correction

Clock timing jitter is another primary noise in the raw measurements [2, 14]. Similarly to laser frequency noise, it exceeds the sensitivity requirement by several orders of magnitude and must be removed in post-processing before gravitational wave signals can be recovered. Unlike laser frequency noise, it couples into the measurement only locally at one spacecraft. Hence, to still allow for a similar differential measurement of the clock noise, the laser fields exchanged between the satellites are modulated with a signal derived from the USOs. From the upper and lower sideband phase measurements at the science interferometer, as written in (8), a correcting term

can be constructed.

Using the phase-domain counterpart of the clock-transfer construction in [14], and neglecting readout and modulation noises initially, we define

$$r_{ji} = \frac{\text{sci}_{ji}^{\text{sb},+}(t) - \text{sci}_{ji}^{\text{c}}(t)}{\nu_{ij}^m} = \mathbf{D}_{ji}q_i(t) - q_j(t). \quad (14)$$

The retained GW phase cancels in this difference because we have approximated the carrier and sideband GW couplings as equal. Keeping their optical-frequency difference would instead add $-H_{ji}$ to r_{ji} ; the cancellation is therefore approximate. The normalization uses the transmitted modulation frequency ν_{ij}^m , giving r_{ji} units of seconds. Equivalently, $r_{ji} = (\text{sci}_{ji}^{\text{sb},+} - \text{sci}_{ji}^{\text{sb},-})/(2\nu_{ij}^m)$ under the same approximations. These variables contain delayed remote clock noise minus local clock noise, matching the structure of laser noise in the science readouts. Delayed combinations of the r variables can therefore be used to construct clock corrections for the TDI observables [14]. The modulation tone must carry the same timing fluctuations that enter the phase measurement. In LISA, pilot-tone correction removes the ADC sampling jitter, making the pilot tone the relevant clock reference. Noise added between this reference and the modulation tone limits the subsequent clock correction. Let $\tau_{ij}^{\text{mod}}(t)$ denote differential timing noise in the tone derivation for MOSA ij , in seconds, and let $\psi_{ij}^{\text{mod}}(t)$ denote additional optical modulation phase noise, in cycles. The excess upper-sideband phase is

$$m_{ij}(t) = \nu_{ij}^m \tau_{ij}^{\text{mod}}(t) + \psi_{ij}^{\text{mod}}(t).$$

Including these terms and readout noise gives

$$\begin{aligned}
r_{ji} &= \mathbf{D}_{ji}q_i - q_j + \frac{\mathbf{D}_{ji}m_{ij} - m_{ji}}{\nu_{ij}^m} \\
&\quad + \frac{N_{ji}^{\text{sb},+} - N_{ji}^{\text{c}}}{\nu_{ij}^m}.
\end{aligned}$$

Modulation and readout noise are kept separate because their transfer functions differ. A source modulation error appears in both local and delayed measurements, while a readout error is added in a particular measurement channel. Independent sources may be uncorrelated, but repeated appearances of the same source must still be combined before calculating its output noise power.

Dividing the modulation phase by its frequency gives $m_{ij}/\nu_{ij}^m = \tau_{ij}^{\text{mod}} + \psi_{ij}^{\text{mod}}/\nu_{ij}^m$. A higher modulation frequency therefore reduces the effect of a fixed phase noise, but leaves a fixed timing noise unchanged. Improving that timing-noise floor requires a quieter tone-generation chain. When the local and transmitted frequencies differ, the local timing contribution retains the factor ν_{ji}^m/ν_{ij}^m . A measurement at one modulation frequency cannot distinguish the two contributions.

The two modulation-frequency synthesis chains can add different excess noises. Local reference-interferometer sideband measurements provide additional observables for correcting the accessible differential

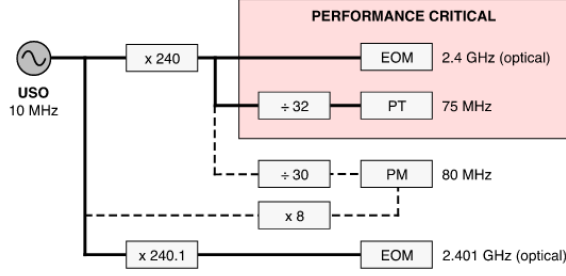


FIG. 2. Clock-generation and distribution scheme highlighting performance-critical paths. A 10 MHz USO signal is multiplied to 2.4 GHz for optical sideband modulation via the EOM, forming the primary clock-transfer tone. Parallel synthesis chains generate the 75 MHz PT used for ADC timing correction and the 80 MHz sampling clock driving the PM. The shaded region marks the performance-critical elements, where clock phase noise must be preserved with high fidelity. A slightly offset 2.401 GHz modulation is generated for the other MOSA to produce MHz-offset sideband-sideband beat notes. The architecture ensures coherent distribution of clock noise into both optical modulation and digital sampling paths, enabling subsequent clock-noise suppression in post-processing.

part [14]. This contribution is absent from the present one-modulator-per-spacecraft testbed.

III. MINILISA DESIGN

A. Motivation for the Present Work

The present work extends the electronic delay-line testbed with optical inputs and independent spacecraft clocks. It combines physical laser noise, sideband-based clock transfer, and hardware delays of several seconds, and uses component measurements to estimate the performance of the combined setup.

The electronic delays place samples acquired several seconds apart in the same interferometric measurement. This lets us test noise cancellation with independently clocked hardware, including changes in the delay caused by the hardware clock itself.

B. Measurement Philosophy

The design provides six directed links between three spacecraft, allowing LISA TDI combinations to be formed within the measurement model described below. We use the familiar Michelson X for the performance analysis; it uses four of these links. The testbed reproduces the laser frequency noise (LFN), clock-noise, and GW couplings of the LISA measurements. The first index of a link denotes the receiving spacecraft (SC); source quantities carry a single spacecraft index, $\delta\phi_i$, q_i , and ν_i^m .

The target carrier phase fluctuations are

$$\delta\phi_{ij}^c(t) = \mathbf{D}_{ij}\delta\phi_j(t) - \delta\phi_i(t) + \delta\phi_{ij}^{\text{GW}}(t) - \alpha_{ij}q_i(t),$$

and the upper-sideband fluctuations are

$$\delta\phi_{ij}^{\text{sb},+}(t) = \mathbf{D}_{ij}\delta\phi_j(t) - \delta\phi_i(t) + \delta\phi_{ij}^{\text{GW}}(t) - (\alpha_{ij} + \nu_j^m)q_i(t) + \nu_j^m\mathbf{D}_{ij}q_j(t).$$

Here α_{ij} is the signed nominal carrier beat frequency. The sideband sampling coefficient is $\gamma_{ij} = \alpha_{ij} + \nu_j^m - \nu_i^m$; combining it with the local modulation phase gives the clock term above. Both channels use the common GW phase approximation of Section II. We call a measurement *LISA-like* when it reproduces these couplings. Modulation, board, and readout noises are added below.

An FPGA delay line supplies the propagation delay and the prescribed Doppler and GW frequency shifts. Heterodyning the optical fields with a common reference laser brings their phase information into the MHz range accessible to the electronics. The reference laser noise becomes part of the effective input laser noise, whose cancellation is discussed below.

The testbed omits the test-mass and reference interferometers, together with test-mass acceleration noise, optical-bench motion, backlink noise, and tilt-to-length coupling. Consequently, the local combinations T_{ji} and ξ_{ji} used to remove bench motion in Section II are unnecessary. Identifying the two bench lasers on each spacecraft with a single source also removes the need for the reference differences z_{ji} . The synthesized links reproduce the three-laser structure of the η observables, but omit the reference-interferometer clock terms proportional to β_{ji} . The clock correction tested here is therefore the reduced case of the LISA construction with those terms absent.

C. Optical Input

Three lasers represent the spacecraft, each with an independent clock driving its modulation and phase measurement. A fourth laser provides the common optical reference. The role of the reference laser is to convert the optical phase fluctuations of each SC laser into electronically processable heterodyne beat notes, which can then be measured by the delay line. This heterodyning step effectively downconverts the optical signals from approximately 281 THz to tens of megahertz, preserving laser frequency noise in the millihertz range of interest. Among the SC lasers, one is frequency-stabilized to a 25 cm ultra-low expansion (ULE) glass cavity, while the reference laser is locked to a 30 cm ULE cavity via the Pound-Drever-Hall method. Both cavities are housed in a vacuum chamber surrounded by three layers of aluminum thermal shielding. The target is a pre-stabilized LFN level of around $30 \text{ Hz}/\sqrt{\text{Hz}}$ in the millihertz band. The remaining two SC lasers are PLL-locked to the synthetic LISA-like beat notes.

Each spacecraft uses an independent rubidium frequency standard as the Moku reference. The Moku supplies both the phase readout and the modulation tone driving the fiber-coupled EOM. We use q_i for the timing fluctuations common to the EOM drive and the phasemeter's sampling reference. Excess noise between these paths enters the modulation and readout terms separately.

The forthcoming modulation measurement will test how faithfully the Moku output carries this common timing noise. If its excess noise is too large, an Rb-referenced signal generator can instead supply the Moku reference and the EOM drive.

The reference laser is unmodulated in the configuration described here, so the spacecraft carrier and sidebands each beat with the reference carrier. The modulation frequency must therefore keep all three electrical beat notes within the delay-line input bandwidth. These input tones carry the optical phase and clock information; the LISA-like carrier-carrier and sideband-sideband readouts are formed by digital subtraction downstream. Modulating the reference laser is a possible extension for higher modulation frequencies, discussed in Section V C. The construction of a two-way arm link is illustrated in Fig. 1(b), alongside a simplified schematic of the corresponding LISA link showing only the functionally relevant elements. Taking the signed photodetector beat phase as SC laser minus reference laser, the carrier signal phase before additive detector noise is

$$\Phi_{PD,i}^c(t) = \nu_{R,i}t + \delta\phi_i^{\text{eff}}(t),$$

where $\nu_{R,i}$ is the signed nominal beat note frequency between the reference laser and the SC i laser, and $\delta\phi_i^{\text{eff}} = \delta\phi_i - \delta\phi_R$ includes their relative laser phase noise. The upper-sideband phase is

$$\begin{aligned} \Phi_{PD,i}^{\text{sb},+}(t) = & (\nu_{R,i} + \nu_i^m)t \\ & + \delta\phi_i^{\text{eff}}(t) + \nu_i^m q_i(t) \\ & + \nu_i^m \tau_i^{\text{mod}}(t) + \psi_i^{\text{mod}}(t). \end{aligned}$$

Here, τ_i^{mod} is timing noise added by the modulation path relative to the common reference q_i , while ψ_i^{mod} is additional modulation phase noise. Noise added only by the sampling path is included in the readout terms below. These are the single-index counterparts of τ_{ij}^{mod} and ψ_{ij}^{mod} in Section II C, with the same dependence on modulation frequency. We write their combined upper-sideband phase noise as

$$m_i(t) = \nu_i^m \tau_i^{\text{mod}}(t) + \psi_i^{\text{mod}}(t).$$

The lower sideband carries the opposite modulation phase, $-\nu_i^m q_i - m_i$.

The photodetector output is then split and distributed across three delay-line boards. One of these boards synthesizes the local LISA-like beat note signals, the other two do so for the other two SC. The signal routing between the boards is illustrated in Fig. 3.

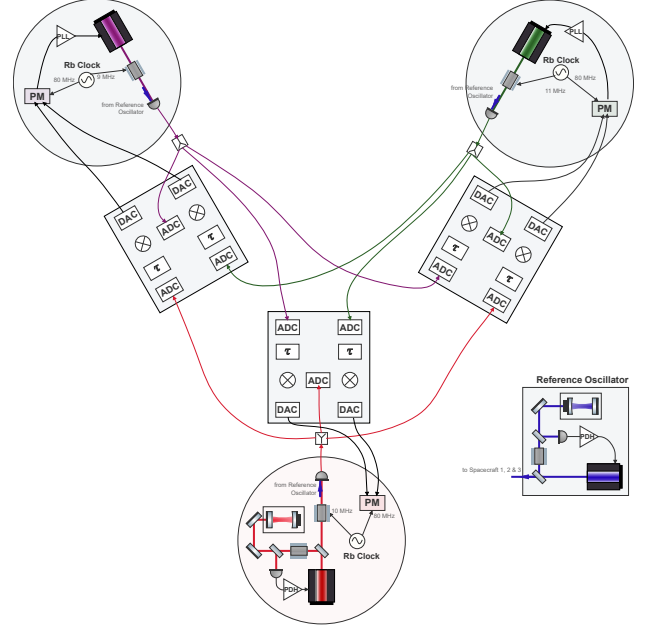


FIG. 3. Planned six-link routing architecture for miniLISA. Each spacecraft has an optical source and an Rb-referenced Moku phasemeter that also supplies its EOM drive; each delay-line board can synthesize two incoming links. The complete routing supports six directed links, whereas the Michelson calculation in Section IV uses links 12, 21, 13 and 31. The spacecraft-1 and reference lasers are stabilized to separate optical cavities; the other spacecraft lasers are phase-locked to synthesized beat notes. The common reference laser supplies the optical heterodyne reference.

Shot noise, intensity noise and detector electronics add fluctuations to the photocurrent, which produce phase errors when the delay-line PLLs track the beat notes. Noise added before the electrical split is shared between boards; ADC and electronics noise added after the split need separate terms. Their phase-noise scales are calculated in Section IV B, and their different propagation is described in Section IV C.

D. The Delay Line

The digital delay line emulates the light-travel time along each directed link. The current firmware always tracks and stores the carrier and both sidebands. At the resulting data rate, the available memory supports a maximum delay of approximately 8.5 s, covering LISA's nominal one-way light-travel time of about 8.3 s.

The implementation uses an AMD Xilinx ZCU208 RF-SoC evaluation board, with integrated ADCs and DACs, FPGA programmable logic, and an embedded processing unit (PS). For each synthesized link, separate DLLs track the carrier and sidebands of the remote and local input signals. The tracked remote frequencies are buffered to impose the arm delay, then combined with the current

local frequencies. The present 125 MHz input-processing rate restricts the optical heterodyne tones to below 62.5 MHz. This is a constraint on the laboratory optical frequency plan, distinct from the frequencies of the synthesized LISA-like outputs.

Orbital delays and GW link responses are computed in advance in Python and uploaded to the PS. During playback, the PS converts the model inputs into fixed-point commands and streams timestamped records to the programmable logic. The buffer supplies the macroscopic light-travel time, while fine delay actuation follows the prescribed arm-length evolution using fractional-delay interpolation. The underlying electronic architecture is described in [9].

The FPGA subtracts the local frequency from the delayed remote frequency and adds the prescribed Doppler and GW offsets. Integrating this frequency produces the output phase for each of the carrier and sideband tones, which are combined at the DAC. Section III E describes the GW scaling. Omitting the nominal frequency evolution, the intended carrier phase is

$$\mathbf{D}_{ij}\delta\phi_j^{\text{eff}} - \delta\phi_i^{\text{eff}} + \delta\phi_{ij}^{\text{GW}}.$$

The performance calculation below uses static delays.

The two ADC inputs and the DAC output share the timing jitter $\epsilon_i(t)$ of the board on spacecraft i . Sampling imprints this jitter on both inputs. The remote contribution then passes through the delay, while subtracting the local signal in the digital mixer reverses the sign of its ADC contribution. The DAC adds jitter at the output beat frequency α_{ij} , giving

$$\begin{aligned}\delta\phi_{\epsilon,ij}^c(t) &= -\nu_{R,j}\mathbf{D}_{ij}\epsilon_i(t) \\ &\quad + \nu_{R,i}\epsilon_i(t) + \alpha_{ij}\epsilon_i(t) \\ &= \nu_{R,j}(1 - \mathbf{D}_{ij})\epsilon_i(t).\end{aligned}$$

Here $\alpha_{ij} = \nu_{R,j} - \nu_{R,i}$, so the local ADC and DAC terms combine to give $\nu_{R,j}\epsilon_i(t)$.

The same cancellation applies to the upper sideband, with the remote input frequency replaced by $\nu_{R,j} + \nu_j^m$:

$$\delta\phi_{\epsilon,ij}^{\text{sb},+}(t) = (\nu_{R,j} + \nu_j^m)(1 - \mathbf{D}_{ij})\epsilon_i(t).$$

Both expressions assume constant nominal frequencies and no additional output frequency offset.

The same board jitter also changes the physical duration of a programmed delay. For a constant command d_{ij} in board time $u_i(t) = t + \epsilon_i(t)$, the input and output times satisfy $u_i(t) - u_i(t_{\text{in}}) = d_{ij}$. To first order,

$$d_{ij}^{\text{phys}}(t) = t - t_{\text{in}} = d_{ij} + \delta d_{ij}^c(t), \quad (15)$$

$$\delta d_{ij}^c(t) = -(1 - \mathbf{D}_{ij})\epsilon_i(t). \quad (16)$$

Thus the physical delay can fluctuate even when its programmed value is known exactly. In LISA, ranging errors leave uncertainty in the light-travel times used for TDI. Here, board timing changes the realized delay itself.

Both effects can leave laser noise if the delays applied in processing do not match the physical delays.

The nominal carrier phase error $-\nu_{R,j}\delta d_{ij}^c$ is already included in the board-jitter term above. The fluctuating delay also acts on the laser phase noise, producing $-\delta d_{ij}^c \mathbf{D}_{ij} \delta\dot{\phi}_j^{\text{eff}}$. This product of timing and laser-frequency noise lies beyond the linear model; it is not included in the static performance estimate.

E. GW Signal Injection

The GW phase defined in Eq. (4) is implemented electronically after the background delay. The required phase on the link from j to i is

$$\delta\phi_{ij}^{\text{GW}}(t) = -\nu_j H_{ij}(t).$$

Since the nominal optical frequency ν_j is constant, the frequency offset supplied to the phase accumulator is

$$\delta\nu_{ij}^{\text{inj}}(t) = \delta\dot{\phi}_{ij}^{\text{GW}}(t) = -\nu_j \dot{H}_{ij}(t). \quad (17)$$

Accumulating this offset reproduces the required GW phase up to an initial phase constant. The scale factor is the represented optical frequency, not the MHz beat note frequency entering the delay-line ADC. Thus the GW signal is generated by adding phase evolution to the reconstructed signal; the hardware delay itself remains d_{ij} and is not additionally perturbed by H_{ij} . This implements the first-order propagation model of Section II.

Each directed-link waveform is evaluated at its reception time t . The propagation and source geometry are already included in H_{ij} , so no additional background delay is applied to the injection waveform after it is generated. Relative timing between the playback streams must reproduce this convention.

The same GW offset is applied to the carrier and both sidebands, following the approximation in Section II. It therefore cancels from their difference used for clock correction while remaining in the science signal.

When the link response is supplied as a fractional-frequency signal $y_{ij} = -\dot{H}_{ij}$, the command is $\delta\nu_{ij}^{\text{inj}} = \nu_j y_{ij}$, using $\nu_j \simeq c/(1064 \text{ nm})$ for the optical scale. The link responses can be computed in Python for a chosen waveform and constellation geometry. Multiple sources can be combined by summing their link responses in the linear regime. The source class is not fixed by the hardware, but the resulting commands must lie within its frequency range, resolution and update bandwidth.

In the delay-line implementation, a signed 32-bit GW word K_{ij} is added on the 64-bit phase-increment scale, giving

$$\delta\nu_{ij}^{\text{inj}} = K_{ij} \frac{f_{\text{clk}}}{2^{64}}.$$

For the nominal 125 MHz accumulator clock, this corresponds to approximately 6.78×10^{-12} Hz per least significant bit and an offset range of ± 0.01455 Hz. The word is

sign-extended to 64 bits without changing its frequency scale and held between timestamped updates. The realized phase accuracy depends on command quantization, update timing, and output phase reconstruction, so command resolution alone does not establish injection fidelity.

F. Readout and Data Processing

The electrical signals produced by the delay line are digitized and tracked by Moku phasemeters, with each spacecraft's phase measurements referenced to its own clock.

The synthesized links can be used directly as η observables. Adding in the modulation, board, and readout noises, the carrier measurement is

$$\eta_{ij}(t) = \mathbf{D}_{ij}\delta\phi_j^{\text{eff}}(t) - \delta\phi_i^{\text{eff}}(t) + \delta\phi_{ij}^{\text{GW}}(t) - \alpha_{ij}q_i(t) + \nu_{R,j}(1 - \mathbf{D}_{ij})\epsilon_i(t) + b_{ij}^c(t),$$

and the corresponding upper-sideband–upper-sideband beat note as

$$\begin{aligned} \eta_{ij}^{\text{sb},+}(t) = & \mathbf{D}_{ij}\delta\phi_j^{\text{eff}}(t) - \delta\phi_i^{\text{eff}}(t) + \delta\phi_{ij}^{\text{GW}}(t) - \gamma_{ij}q_i(t) \\ & + \nu_j^m \mathbf{D}_{ij}q_j(t) - \nu_i^m q_i(t) \\ & + \mathbf{D}_{ij}[\nu_j^m \tau_j^{\text{mod}}(t) + \psi_j^{\text{mod}}(t)] \\ & - \nu_i^m \tau_i^{\text{mod}}(t) - \psi_i^{\text{mod}}(t) \\ & + (\nu_{R,j} + \nu_j^m)(1 - \mathbf{D}_{ij})\epsilon_i(t) + b_{ij}^{\text{sb},+}(t). \end{aligned}$$

The terms proportional to q_i describe timing common to the modulation and phasemeter reference, while those proportional to ϵ_i come from the delay-line board.

The additive phase noise b_{ij}^a combines optical readout, residual board, and final-phasemeter noise, in cycles; timing and modulation noise are included explicitly. To retain the noise shared by copies of a photodetector signal, write

$$b_{ij}^a = \mathbf{D}_{ij}p_j^a - p_i^a + e_{ij}^a, \quad (18)$$

where p_i^a is the equivalent input phase noise in channel a before the split, and e_{ij}^a contains noise added by the individual electronic paths. This expression assumes matched tracking responses. Any mismatch must be included in the transfer functions rather than treating copies of p_i^a as independent sources.

The effective laser phases $\delta\phi_i^{\text{eff}}$ enter with the same difference structure as the LISA laser phases. TDI therefore suppresses both the spacecraft and reference laser noise.

Finally, the upper-sideband-minus-carrier variable

takes the form

$$\begin{aligned} r_{ij} = & \frac{\eta_{ij}^{\text{sb},+} - \eta_{ij}}{\nu_j^m} \\ = & \mathbf{D}_{ij}q_j - q_i + \frac{\mathbf{D}_{ij}m_j - m_i}{\nu_j^m} \\ & + (1 - \mathbf{D}_{ij})\epsilon_i + \frac{b_{ij}^{\text{sb},+} - b_{ij}^c}{\nu_j^m}. \end{aligned}$$

The common GW phase cancels, leaving the clock-transfer signal together with modulation, board and readout noise.

The board jitters are strongly correlated across the setup. Writing $\epsilon_i = \epsilon_{\text{com}} + \Delta\epsilon_i$ separates their shared fluctuation from the differential jitter. In the constant-frequency model above, the common term can be absorbed into the effective laser phases and clock errors as $\delta\phi_i^{\text{eff}} - \nu_{R,i}\epsilon_{\text{com}}$ and $q_i - \epsilon_{\text{com}}$. It therefore cancels under TDI together with clock correction. Differential board jitter remains and sets the relevant timing residual. A separate timing measurement could be used to subtract this residual if needed; strong common-mode cancellation means that such an extra correction is not automatically required.

Table I summarizes the noise sources shared with LISA, those omitted from the testbed, and those introduced by its laboratory implementation. The readout predictions are derived in Section IV B.

TABLE I. Noise sources in LISA and miniLISA, including the terms in Eqs. (7), (8), (10), and (11), and their treatment in the present analysis. “Not reproduced” and “omitted” refer to the intended measurement model; they do not imply a noise-free laboratory. Readout numbers compare LISA science-interferometer allocations with the detector-noise predictions of Section IV B for 4 mW per beam; they are not measured total sensitivities.

Noise source	LISA	miniLISA	Treatment / relevance
Present in both LISA and miniLISA			
Laser frequency noise	Present	Present	Suppressed by TDI with matched delays; ideal cancellation in this estimate.
Measurement-clock jitter q_i	Present	Present	Sideband-based post-processing correction; residual tone-generation noise remains.
Modulation noise	Present	Present	Limits clock correction. Kept as source noise with its own transfer function.
Science readout noise N_{ij}^c, N_{ij}^{sb}	Weak received light; shot noise important	Bright beams; shared detector inputs	Shared carrier noise cancels in ideal TDI; channel differences enter clock correction.
Carrier / sideband readout ASD	Allocations: $9 \text{ pm}/\sqrt{\text{Hz}}$ / $600 \mu\text{rad}/\sqrt{\text{Hz}}$	Model: $0.0036 \text{ pm}/\sqrt{\text{Hz}}$ / $0.0778 \mu\text{rad}/\sqrt{\text{Hz}}$	LISA plateau values versus an illustrative linear detector model; assumed optical power and ideal electronics.
ADC and phasemeter noise	Present	Present	Ideal ADC quantization included; sampling jitter and residual electronics treated separately.
Delay mismatch / processing residuals	Ranging and processing errors	Board-induced delay fluctuations and processing errors	Track effective delays; Only matched static delays are used in the present estimate.
LISA sources not reproduced in miniLISA			
Test-mass displacement δ_{ij}	Includes acceleration noise	Not reproduced	Survives in LISA observables; acceleration noise enters only the comparison curve here.
Optical-bench displacement Δ_{ij}	Present	Not reproduced as a separate degree of freedom	Removed by the LISA test-mass/reference combinations forming ξ_{ij} .
Backlink fibre noise μ_i	Present	No inter-bench backlink	The reciprocal part cancels in the local combinations; nonreciprocal noise would remain.
Test-mass readout noise N_{ij}^{tm}	Present	No test-mass interferometer	Enters LISA through the bench-motion correction; no corresponding measurement in miniLISA.
Reference readout noise N_{ij}^{ref}	Present	No local reference interferometer	Enters the LISA local combinations; the laboratory reference laser is a different component.
Tilt-to-length coupling	Present	Not explicitly modelled	Laboratory alignment and path noise can still enter the measured residual.
Additional sources in miniLISA			
Shared reference-laser noise	No laboratory reference laser	Present	Part of the effective source phase; suppressed by TDI in the ideal model.
Delay-board timing jitter ϵ_i	No electronic arm-delay board	Present	Shared jitter cancels in TDI with clock correction; differential jitter can remain.
Residual delay-board electronics	No electronic arm-delay board	Present	Measured baseline included as additive noise; not assumed to cancel.

IV. PERFORMANCE MODELLING

We estimate the noise remaining after TDI and clock correction from the measured delay-line noise, the modulation characterization and a prospective optical readout calculation. The calculation uses static delays and ideal cancellation of the common laser and clock noises.

A. Noise Inputs and Comparison Reference

For laser frequency noise we adopt the LISA pre-stabilization reference [3],

$$A_\nu(f) = 30 \frac{\text{Hz}}{\sqrt{\text{Hz}}} \sqrt{1 + \left(\frac{2 \text{ mHz}}{f}\right)^4}. \quad (19)$$

Its equivalent phase ASD is $A_\phi = A_\nu/(2\pi f)$.

The clock estimate uses the phase-noise model of a 10 MHz Rb standard. Its single-sideband phase noise is anchored at $\mathcal{L}(10\text{ Hz}) = -130\text{ dBc/Hz}$, consistent with the specifications in [15]. Below this point, the phase PSD is extrapolated as f^{-1} down to 1 Hz, f^{-2} down to 10 mHz, and f^{-3} below 10 mHz. The corresponding timing ASD is

$$A_q(f) = \frac{\sqrt{2 \cdot 10^{\mathcal{L}(f)/10}}}{2\pi \times 10\text{ MHz}}. \quad (20)$$

This gives approximately $2.25\text{ ps}/\sqrt{\text{Hz}}$ at 10 mHz and $71\text{ ps}/\sqrt{\text{Hz}}$ at 1 mHz. It is an estimate from the assumed low-frequency continuation, rather than a measurement of the laboratory clocks. The common timing noise couples to a carrier link as $|\alpha_{ij}|A_q$ and is removed by the ideal clock correction used here.

The electronic input is the measured delay-line phase-noise floor, including processing and electronics and possibly the readout phasemeter's ADC. The carrier and sideband measurements give the same floor, so we assign the same PSD to each electronic readout term. The terms are taken to be independent between links and channels, while repeated uses of a given readout remain coherent.

The modulation characterization uses half the upper-minus-lower sideband phase of a local optical beat, with a common modulation and measurement reference. The present spectrum is used as an estimate of source modulation phase noise m_i . The planned repeat measurement will update this input and test the proposed Moku-generated modulation tone.

Figure 4 shows these component inputs in cycles/ $\sqrt{\text{Hz}}$. The measured spectra cover the plotted 0.25 mHz–1 Hz band without extrapolation.

For comparison, we use the single-link reference adopted in the component analysis. With $\lambda = 1064\text{ nm}$, its equivalent phase ASDs are

$$A_{\text{OMS}}(f) = \frac{15 \times 10^{-12}\text{ m}}{\lambda} \sqrt{1 + \left(\frac{2\text{ mHz}}{f}\right)^4}, \quad (21)$$

$$A_{\text{acc}}(f) = \frac{3 \times 10^{-15}\text{ m s}^{-2}}{(2\pi f)^2 \lambda} \sqrt{1 + \left(\frac{0.4\text{ mHz}}{f}\right)^2} \times \sqrt{1 + \left(\frac{f}{8\text{ mHz}}\right)^4}. \quad (22)$$

The reference is $A_{\text{link}} = (A_{\text{OMS}}^2 + A_{\text{acc}}^2)^{1/2}$. Acceleration noise is included here as a comparison quantity; it is not an experimentally present test-mass noise in the current testbed. The component spectra are displayed directly, without applying a TDI transfer function. Consequently, this figure compares input noise scales and does not establish a gravitational-wave sensitivity.

B. Predicted Carrier and Sideband Readout Noise

For the readout calculation, we consider one phase-modulated laser interfering with an unmodulated reference laser in a prospective Newport PIN-detector readout. Each beam delivers 4 mW at the detector, after any combining losses, and the phase-modulation depth is $\mu_{\text{EOM}} = 0.53\text{ rad}$. This operating mode provides an explicit carrier-carrier and sideband-carrier readout estimate. The optional configuration with both lasers modulated requires the corresponding sideband-sideband beat amplitudes to be substituted; the present numerical values should not be transferred to that mode without this adjustment.

For optical powers P_{SC} and P_{ref} , responsivity $\mathcal{R}$, and power-overlap efficiency η_{het} , the peak current of beat order n is

$$I_{n,\text{pk}} = 2\mathcal{R}\sqrt{\eta_{\text{het}}P_{\text{SC}}P_{\text{ref}}}|J_n(\mu_{\text{EOM}})|. \quad (23)$$

The carrier uses $n = 0$, and either first-order sideband uses $|n| = 1$. Here $J_0(0.53) = 0.9310$ and $J_1(0.53) = 0.2558$. The modulated beam therefore contains 3.47 mW in its carrier and 0.262 mW in each first-order sideband. Using $\mathcal{R} = 0.6\text{ A/W}$ and $\eta_{\text{het}} = 0.9$ gives peak beat currents of 4.24 mA and 1.16 mA, respectively.

We include shot noise, dark-current shot noise, Johnson noise, relative intensity noise (RIN), and ADC quantization noise. Their one-sided current ASDs are modelled as

$$i_{\text{shot}} = \sqrt{2e[\mathcal{R}(P_{\text{SC}} + P_{\text{ref}}) + I_d]}, \quad (24)$$

$$i_{\text{J}} = \sqrt{4k_{\text{B}}T/R_{\text{L}}}, \quad (25)$$

$$i_{\text{RIN}} = \mathcal{R}(P_{\text{SC}} + P_{\text{ref}})A_{\text{RIN}}, \quad (26)$$

$$i_{\text{ADC}} = \frac{V_{\text{FS}}}{2^{B_{\text{ADC}}}G\sqrt{6f_s}}. \quad (27)$$

The assumed linear-receiver parameters are $I_d = 100\text{ nA}$, $T = 300\text{ K}$, $R_{\text{L}} = 50\ \Omega$, $G = 50/\sqrt{3}\text{ V/A}$, $V_{\text{FS}} = 1\text{ V}$, $B_{\text{ADC}} = 14$ and $f_s = 2\text{ GHz}$. The RIN ASD is $A_{\text{RIN}} = 0.9 \times 10^{-8}/\sqrt{\text{Hz}}$. The RIN expression retains a conservative coherent sum of the two beams' intensity fluctuations. It represents additive noise around the RF beat frequencies; low-frequency amplitude modulation is not assumed to couple directly to phase. The ADC term uses the physical conversion rate $f_s = 2\text{ GS/s}$, before the 125 MHz FPGA processing described in Section III D. It assumes white quantization noise and appropriate antialias filtering; a decimated output rate is not a substitute for f_s in this estimate.

The gain assumes a 50 V/A detector output followed by an ideal matched three-way electrical split. The split attenuates the signal and detector noise equally, while increasing the input-referred ADC contribution. At 4 mW per beam, the mean and maximum instantaneous optical powers are 8 and 15.6 mW, respectively, below the detector's listed 20 mW maximum [16]. After DC removal, the ideal phase-modulated beat has a peak-to-peak voltage of

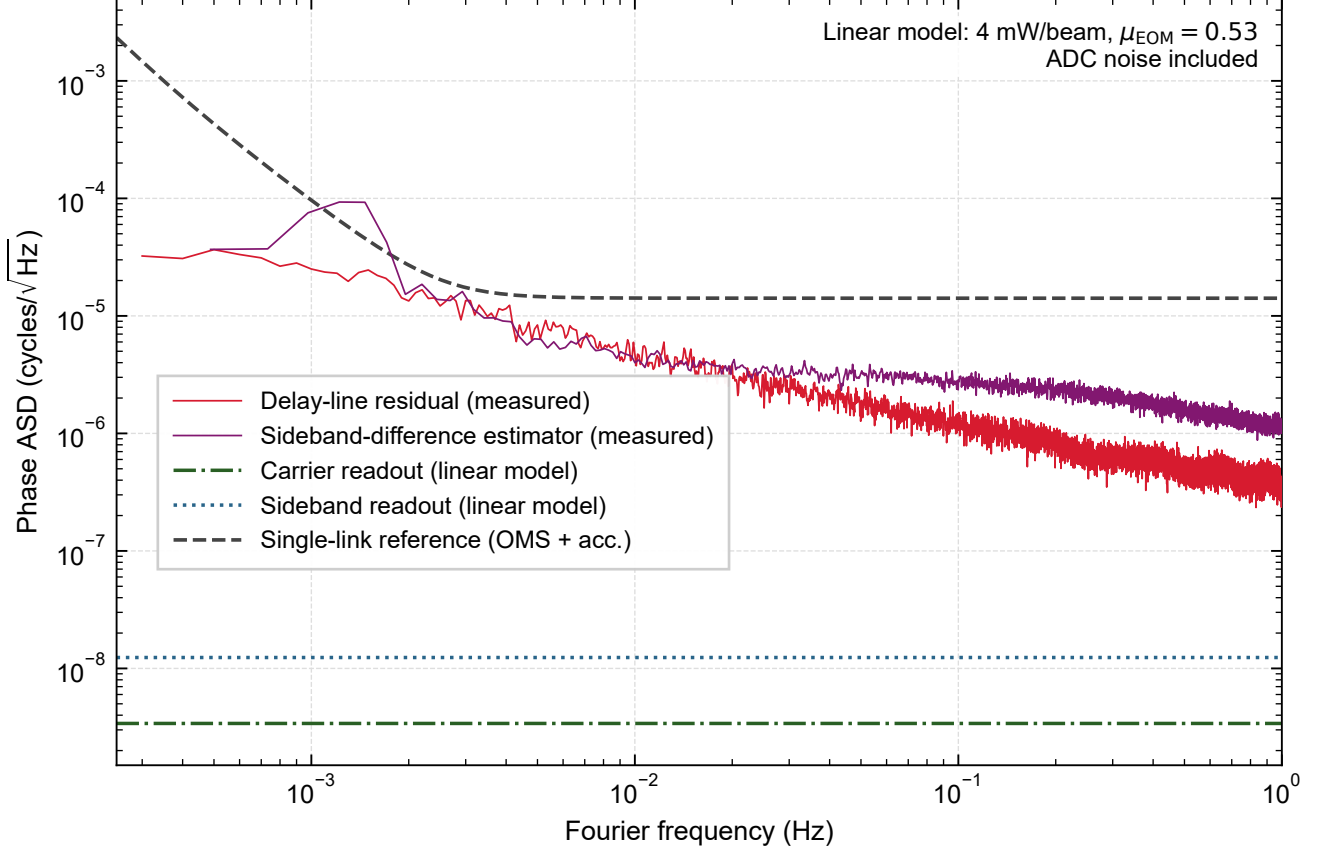


FIG. 4. Component phase-noise amplitude spectral densities compared with the single-link reference formed from optical metrology and acceleration noise. Solid curves show the measured delay-line residual and modulation phase-noise estimator; measured Fourier bins are plotted without smoothing. The illustrative linear-receiver carrier and first-order sideband readout noises include shot noise, dark-current shot noise, Johnson noise, relative intensity noise and ADC quantization noise. The optical model assumes one phase-modulated beam and one unmodulated reference beam, each delivering 4 mW at the detector, with modulation depth 0.53 rad. The detector responsivity is 0.6 A/W, heterodyne efficiency is 0.9, and the termination is 50 Ω at 300 K. The ADC model uses a 1 V full-scale span, 14 bits and 2 GS/s, with an ideal three-way electrical split. The assumed white RIN is $0.9 \times 10^{-8}/\sqrt{\text{Hz}}$, summed coherently between the beams. The model uses (35, 36, 34) MHz modulation and electrical carrier frequencies of 15.0, 10.9 and 21.6 MHz; the readout prediction assumes white noise across the corresponding RF bands. All curves are shown before propagation through TDI and clock correction; the measured electronic residual is shown as characterized and is not deconvolved into individual board noise sources. The comparison is not a measurement of the complete testbed sensitivity.

approximately 0.263 V at each ADC. These assumptions provide headroom for the prospective readout; additional RF losses are not included.

For additive noise that is locally white around the selected beat, demodulation combines noise from both RF sidebands. The equivalent phase ASD of contribution k is

$$A_{\phi,n}^{(k)} = \frac{\sqrt{2} i_k}{2\pi I_{n,\text{pk}}}. \quad (28)$$

The factor 2π converts radians to cycles. No integration over the PLL bandwidth is applied to these spectral densities. Summing the component PSDs gives carrier and first-sideband readout ASDs of 3.40×10^{-9} and $1.24 \times$

10^{-8} cycles/ $\sqrt{\text{Hz}}$, respectively, equivalent to 21.4 and 77.8 nrad/ $\sqrt{\text{Hz}}$. Their ratio is $J_0(\mu_{\text{EOM}})/|J_1(\mu_{\text{EOM}})| \simeq 3.64$. These are predictions for the stated optical powers and white-noise assumptions, not measurements of the complete readout chain.

The signal levels explain why this readout operates differently from LISA's. The 4 mW laboratory beams produce strong carrier and sideband beats, whereas LISA must measure light received after propagation over millions of kilometres. miniLISA therefore has a much lower predicted detector readout noise than the LISA science-interferometer allocations, as summarized in Table I.

The testbed does not need to reproduce LISA's weak-light, shot-noise-limited readout to test laser- and clock-

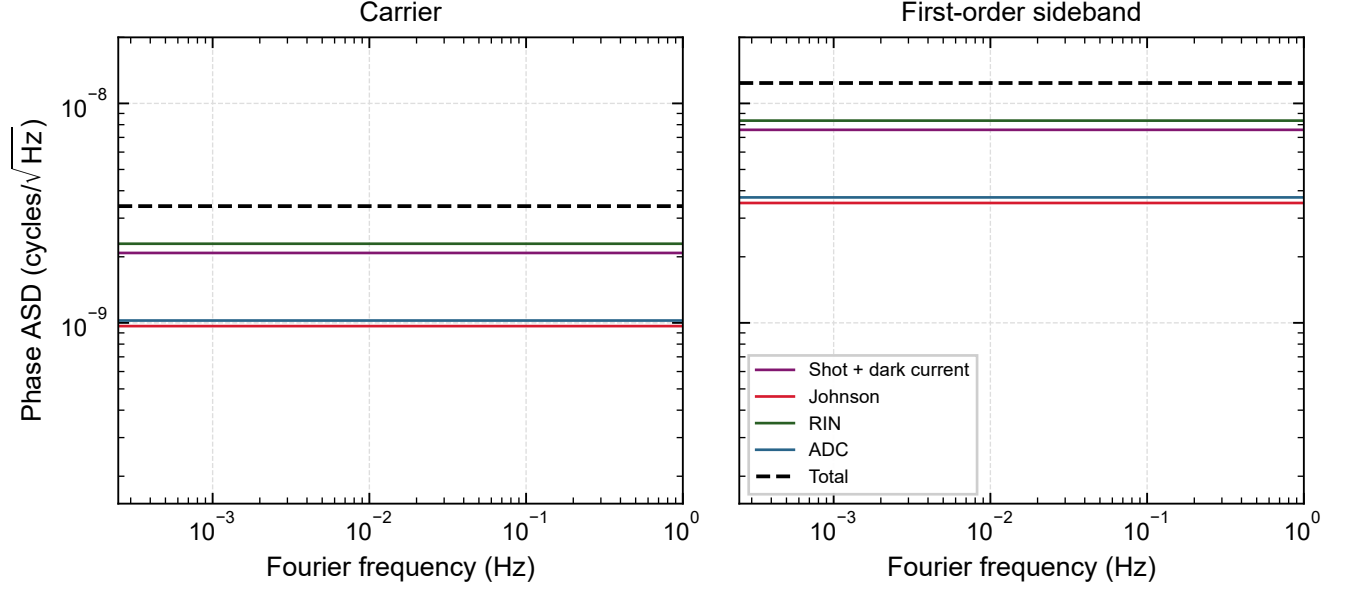


FIG. 5. Illustrative linear-receiver carrier and first-order sideband readout noise for one modulated and one unmodulated beam, each delivering 4 mW at the detector, with modulation depth 0.53 rad. The two panels share a common vertical scale. Shot noise includes dark-current shot noise; the ADC contribution is the white quantization estimate. The dashed curves are the quadrature totals. The beat amplitudes in Eq. (23) set the different phase-noise levels. The ADC estimate includes an ideal three-way electrical split.

noise suppression. A lower readout floor makes smaller residuals from the delay and clock-transfer chains accessible. Shot noise is included in the calculation. The measured electronic residual and sideband-difference estimator exceed the calculated input readout noise; their contributions to the corrected observable depend on the source propagation discussed below. Thus the laboratory signal powers are chosen to support the noise-suppression tests, rather than to reproduce the mission's photon budget.

The LISA allocations (IRD2-IDS-04010 and IRD2-IDS-04580) increase towards low frequencies as $\sqrt{1 + (f_k/f)^4}$, with $f_k = 2$ mHz for the carrier and 0.7 mHz for the sidebands. They refer to individual readouts and are distinct from the total OMS-plus-acceleration comparison curve used above.

These input noise levels do not by themselves specify the output noise after TDI. In particular, detector noise shared by the split input paths and noise added independently by each board have different transfer functions.

C. Propagation Through TDI and Clock Correction

The propagation calculation uses the four links of Michelson X , static delays and constant nominal frequencies. The illustrative frequency plan is $\nu_{R,i} = (15.0, 10.9, 21.6)$ MHz and $\nu_i^m = (35, 36, 34)$ MHz, giving $(\alpha_{12}, \alpha_{21}, \alpha_{13}, \alpha_{31}) = (-4.1, 4.1, 6.6, -6.6)$ MHz. The

nominal one-way delay is $L = 2.5 \times 10^9$ m/c $\simeq 8.339$ s. These are example operating frequencies. The small modulation-frequency offsets separate the synthesized carrier and sideband beats; the laboratory readout is not constrained to the representative LISA 5–25 MHz band. Writing $z_{ij} = \exp(-2\pi i f d_{ij})$, $A = z_{12} z_{21}$ and $B = z_{13} z_{31}$, the static Michelson defined above becomes

$$X = (1 - A)(\eta_{13} + z_{13}\eta_{31}) - (1 - B)(\eta_{12} + z_{12}\eta_{21}). \quad (29)$$

Let P_{ij} denote its four link coefficients and K_{ij} the clock-correction coefficients, chosen so that

$$X_c = \sum_{ij} (P_{ij}\eta_{ij} + K_{ij}r_{ij}) \quad (30)$$

has zero coefficient for each q_i . The sum runs over links 12, 21, 13 and 31, using the upper-sideband-minus-carrier definition of r_{ij} given above. The static second-generation convention used in the calculation is $X_{2c} = (1 - AB)X_c$; this extra filter must multiply both the science and correction weights. Laser and clock cancellation can be checked algebraically for unequal static delays. The calculation is restricted to static arms.

For four independent link noises with the same ASD A_e , the uncorrected TDI output is

$$A_{X,e} = A_e \sqrt{\sum_{ij} |P_{ij}|^2}. \quad (31)$$

Four unit-weight terms would give $\sqrt{4}A_e = 2A_e$. TDI also contains delayed differences, so its weights depend on Fourier frequency. For equal delays d , the static $X_2 = (1 - AB)X$ convention gives $A_{X_2,e} = 8|\sin(2\pi fd)\sin(4\pi fd)|A_e$ before clock correction.

For noise added to individual links, repeated appearances of one readout must be combined coherently. In particular, for the first-generation weights the coefficients are

$$T_{b_{ij}^c} = P_{ij} - \frac{K_{ij}}{\nu_j^m}, \quad T_{b_{ij}^{sb,+}} = \frac{K_{ij}}{\nu_j^m}. \quad (32)$$

The two appearances of a given carrier noise are therefore combined before taking a squared magnitude. These coefficients apply directly to e_{ij}^a in Eq. (18), not to independent copies of a shared detector noise.

For matched paths, the shared carrier term $\mathbf{D}_{ij}p_j^c - p_i^c$ has the laser-noise structure and cancels in carrier TDI. However, define the detector channel difference $\chi_i = p_i^{sb,+} - p_i^c$. It contributes to the clock-transfer variable as

$$r_{ij}^{\text{det}} = \frac{z_{ij}\chi_j - \chi_i}{\nu_j^m}, \quad (33)$$

and hence to the corrected observable as

$$X_{2c}^{\text{det}} = (1 - AB) \sum_{ij} K_{ij} \frac{z_{ij}\chi_j - \chi_i}{\nu_j^m}. \quad (34)$$

For the estimate, detector noise in the carrier and sideband channels is taken to be independent, so $S_{\chi_i} = S_{p_i^{sb,+}} + S_{p_i^c}$. Copies of each detector channel shared between boards retain the same noise realization.

For independent physical sources n_a , the output PSD is

$$S_{X_{2c}}(f) = \sum_a |T_a(f)|^2 S_a(f). \quad (35)$$

The measured electronic spectrum is propagated with the carrier and sideband coefficients above. The predicted detector terms use Eq. (34). ADC quantization is shown in the input readout estimate but is not added separately to the total, where electronics are represented by the measured delay-line floor.

True source modulation noise survives clock correction. At fixed nominal beat frequencies, increasing all ν_i^m by a common factor s reduces its transfer coefficients by $1/s$. The output modulation ASD therefore decreases by $1/s$ for fixed direct phase noise. For fixed timing noise, the input modulation phase ASD increases by s , leaving the residual unchanged. These scaling statements apply to the identified physical source, not automatically to the full measured sideband-difference spectrum. Likewise, a small raw TDI ASD at low Fourier frequency partly reflects the differencing response of the observable, which also acts on a gravitational-wave signal; it is not by itself an improvement in sensitivity.

D. Estimated Residual Noise

Figure 6 combines the electronic, modulation and detector contributions after static second-generation TDI and clock correction. Each physical source is multiplied by its transfer function before the output PSDs are summed. The calculation uses four independent electronic link readouts per channel and three shared detector inputs, with the assumptions given above. The detector contribution is small compared with the measured electronic and modulation inputs, consistent with the use of bright laboratory beams rather than a weak-light readout.

The estimate uses the current modulation spectrum, which will be replaced after the planned repeat measurement. It describes the included component noises under ideal primary-noise cancellation, rather than a measured sensitivity of the assembled optical testbed.

Board timing also changes the physical delay, even when the programmed delay is known exactly. For constant d_{ij} , Eq. (16) gives

$$A_{\delta d_{ij}^c}(f) = 2|\sin(\pi f d_{ij})|A_{\epsilon_i}(f). \quad (36)$$

The common board timing cancels in the linear model discussed above; differential timing can remain. Its nominal beat-frequency coupling is already included in the ϵ_i terms and is not a separate independent ranging noise. No additional timing or delay-error floor is assigned in the present estimate.

Under the common carrier/sideband GW approximation, the injected signal in this static observable is

$$X_{2c}^{\text{GW}} = (1 - AB) \sum_{ij} P_{ij} \delta \phi_{ij}^{\text{GW}}. \quad (37)$$

This provides the amplitude and phase prediction for a subsequent hardware injection test.

V. FUTURE EXTENSIONS

A. Injection of Secondary Noise

Test-mass displacement can be emulated by adding the corresponding phase terms M_{ij} from Section II to the synthesized links. Each simulated test mass must use the same displacement realization wherever it occurs, including the delayed occurrence in the reverse link. Independent noise added to every link would not reproduce these correlations. For a specified acceleration spectrum, the displacement ASD is obtained by dividing by $(2\pi f)^2$ over the analysis band. This injection would test the secondary-noise transfer functions as well as primary-noise suppression.

Independent phase perturbations can also be added after link synthesis to test a prescribed LISA-like readout-noise model. Lower optical power would test weak-light

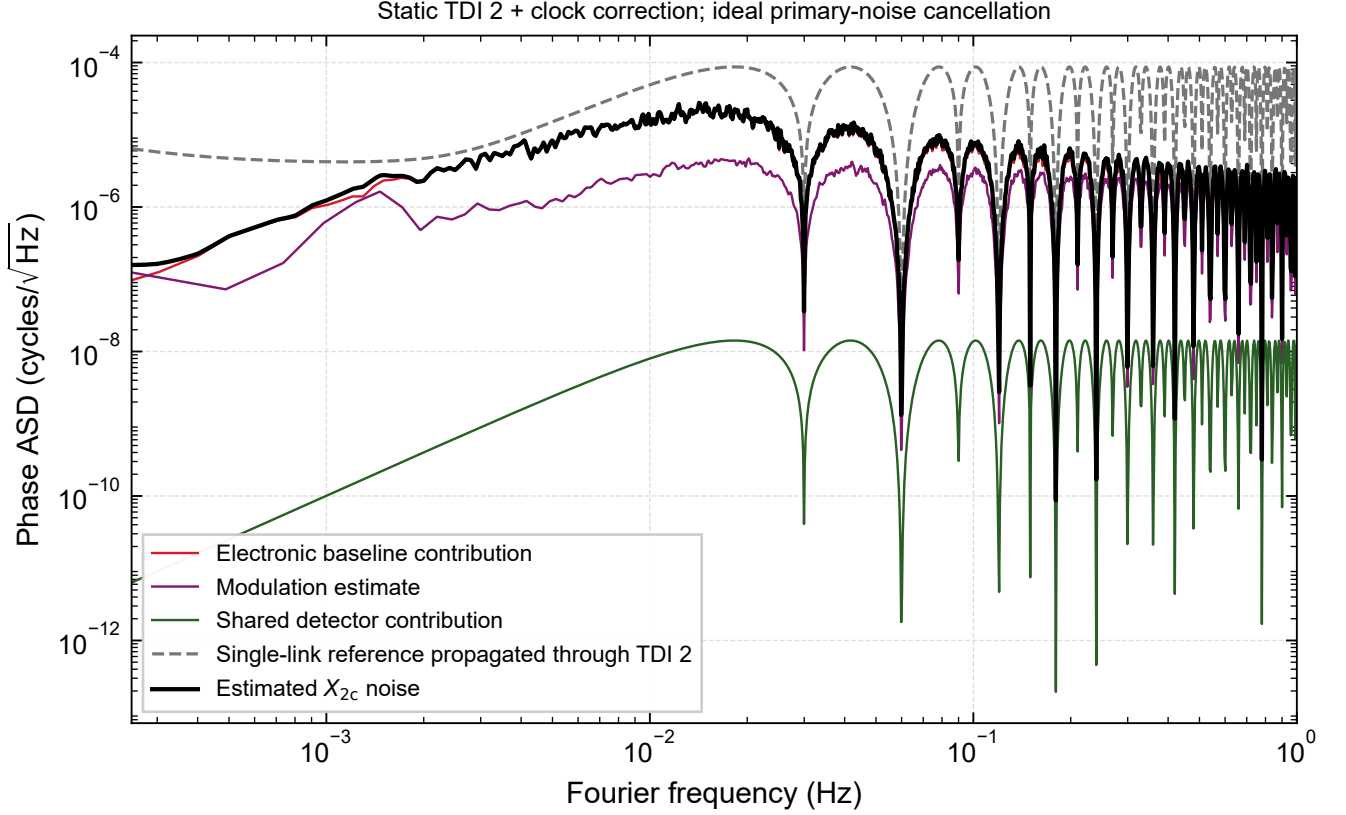


FIG. 6. Estimated phase-noise ASD of the clock-corrected static Michelson observable X_{2c} . The black curve sums the electronic, modulation and shared-detector contributions in quadrature. The calculation uses four links with 8.339 s delays, modulation frequencies (35, 36, 34) MHz, and the optical operating point of Section IV B. ADC noise is represented by the measured electronic baseline. The dashed comparison curve treats the single-link OMS-plus-acceleration reference as independent equivalent link noises propagated through the same TDI observable. Laser and common clock noise cancel ideally in this static estimate.

tracking and readout, but would not by itself remove the correlations introduced by splitting one detector output across boards.

B. Time-Dependent Delays

The six-link design supports TDI combinations beyond the four-link Michelson considered here. Orbit-derived delays and Doppler shifts would extend the present static analysis to arm flexing and second-generation TDI. For frequency data, the delayed phase derivative contains the factor $1 - \dot{d}_{ij}$, which is unity in the static calculation [17].

C. Changes to the Clock-Transfer and Locking Configuration

Increasing the modulation frequency could reduce the effect of fixed excess phase noise in clock transfer, but

only after the measured estimator has been separated into modulation, timing and readout contributions. A GHz implementation would also require a compatible optical and RF frequency plan. Modulating the reference laser would allow corresponding optical sidebands to beat within the electronic input band.

A second optical source and modulator on each spacecraft, together with reference interferometers and backlinks, would reproduce the local $\xi/z/\eta$ reduction and the β clock couplings omitted here [12]. Distinct synthesis chains and a local sideband comparison would also allow tests of differential tone-generation noise.

VI. CONCLUSION

We have presented the miniLISA design, which extends an electronic delay-line testbed with optical sources and independently referenced spacecraft measurements. Building on hardware demonstrations with electronic

propagation delays [6, 9], it is intended to test laser-noise suppression, a reduced clock-transfer model, and GW signal recovery within the same measurement chain.

The estimate combines the measured electronic and modulation spectra with a prospective detector readout model, using the appropriate source and link transfer functions. Bright laboratory beams keep the predicted photodetector readout noise below the LISA science-interferometer allocations. Board timing introduces an additional coupling, including fluctuations of the effective delay, whose residual depends on differential board

jitter and the delays used in processing. This component estimate provides a reference for the performance of the assembled optical testbed.

The next steps are to repeat the modulation-noise measurement, assemble the optical readout, and test noise suppression and GW recovery with the six-link setup. Varying delays and higher modulation frequencies would extend these tests. Higher modulation frequencies reduce the effect of fixed phase noise in clock transfer, while differential timing noise requires improvements to the clock-distribution and modulation chains themselves.

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